

Automatic Recognition of Blooming Flowers

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Abstract

This paper describes an automatic method for recognizing a blooming flower based on a photograph taken with a digital camera in natural scene. The problem of identifying an object against the background is known to be difficult. In this paper, we employ a photograph where the object (a blooming flower) is focused but the background is defocused. For extracting a flower region, we propose a new method that extracts a boundary by selecting a route with minimizing a sum of the local cost divided by the route length. Experiments were conducted for 600 pictures (20 pictures each for 30 species). A successful boundary extraction rate of 97% and a flower recognition rate of 90% were obtained.

1. Introduction

Blooming flowers often attract our attention but we rarely know their names. In this paper we describe a method for naming or recognizing a flower based upon a photograph. The extraction of an object of interest from the background consisting of various objects is an essential step for recognition but is known to be a difficult task. Our previous work[1] used an approach to avoid the difficulty by physically picking up a flower and placing it on a black sheet before taking a photograph. This method is undesirable from an environmental viewpoint in addition to some laborious work involved prior to photographing. In this paper, we employ a photograph that includes a focused object with defocused background. This kind of picture can be easily taken with a digital camera with macro capability.

For recognition of natural objects, Nagy and Zou[2] proposed an interactive recognition system for 10 species of flowers with 10 samples each. Das et al.[3] similarly presented an interactive extraction system

for recognition of flowers based on color. Nakamura et al.[4] proposed a content-based retrieval system for flowers based upon the color distribution of flowers. Im et al.[5] and Sekita et al.[6] worked on the recognition of maple trees from the leaf shape alone. By and large, there have been no works that are able to automatically extract an object of interest on a photograph taken in natural scene.

2. Extraction Method

Figure 1 shows two examples of input photographs. The object, a flower, should be placed on the center. The flower should be well focused and the background reasonably defocused. This kind of picture is easily obtainable with a digital camera with macro capability (e.g. OLYMPUS C-2100).



Figure 1: Input photographs.

2.1. Automatic Extraction

For extracting an object boundary, there is a well-known method by Mortensen and Barrett[7] called Intelligent Scissors (IS). It is a manual method to draw a boundary based upon a number of manually selected points on the visually identified boundary, providing a route that minimizes the sum of local costs. It is fast because the principle of dynamic programming is employed. However, the difficulty is that it requires many manually selected points particularly for an object with a complicated boundary. The number of points required is about 42 and 8 for the left and right picture shown in Figure 1, respectively. (Usually a smaller number of points for smoother boundaries.) In order to deal with flowers with complex boundaries, we proposed a new route search method called the normal-

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ized cost (NC) method[8] which searches a boundary with minimizing the normalized cost i.e. the sum of local costs divided by the route length. With this NC method, entry of one point on the boundary is all the required. But this is as yet a manual method. In the following, we extend this NC method to an automatic one.

On a photographic image, five points are set: P_0 at the center and P_1, P_2, P_3 and P_4 at the middle of four sides as shown in Figure 2. Then four profiles of local costs along the straight lines from P_0 to all of the four middle points are computed. (Note here that the local cost is a weighted sum of Laplacian zero-crossing, gradient magnitude, and route direction as the IS method[7]). Then on each profile, three local minimum points are identified. Consequently a total of 12 points are found. Figure 4 shows four profiles observed from the picture shown in Figure 3. Figure 4 depicts 12 local minimum points. These 12 points are the starting points of our NC method.

Then the NC method is run for each of the 12 points as the starting point. The resulting 12 boundaries or routes are shown in Table 1 and in Figure 5. Some are correct but the others are incorrect boundaries. In order to pick correct boundaries, the following screening process is carried out:

- (1) Remove any small boundary whose area is less than 3% of picture size.
- (2) Remove any boundary that does not contain the center point.
- (3) Among the remaining boundaries, select the boundary yielding the smallest average cost.

For the case of Figure 5, the boundary 1 is the final boundary selected.



Figure 2: Location of five points. Figure 3: Detected 12 points.

Table 1: Resulting 12 regions of Figure 5.

No.	area [pixels]	normalized cost	No.	area [pixels]	normalized cost
1	91935	32.4	7	91935	32.4
2	92030	32.7	8	91936	32.4
3	8801	56.5	9	62234	36.5
4	92030	32.8	10	91937	32.4
5	8766	57.4	11	107	198.8
6	923	63.3	12	8871	121.3

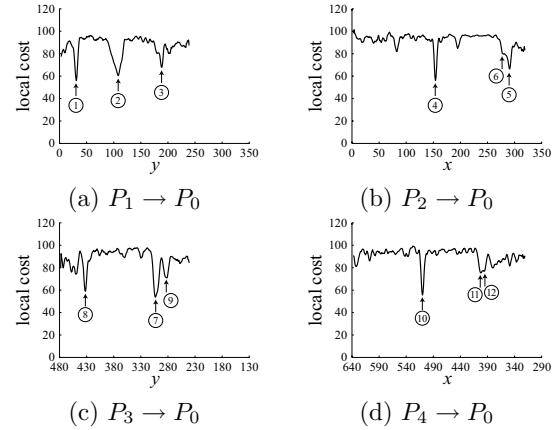


Figure 4: Four cost profiles.

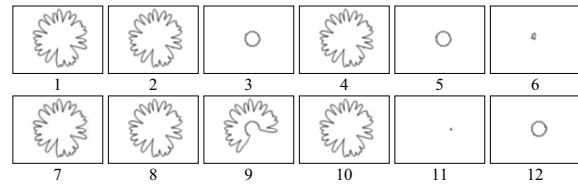


Figure 5: 12 extracted regions.

2.2. Interactive Extraction

Before running the recognition algorithm (to be described in the next section), the resulting boundary is presented to a user for verification. For a small percentage of pictures (17 pictures out of 600 in our experiment), the above automatic method failed to provide correct boundaries from various reasons. Figure 6 is an example. The reason for this case is that a clear edge situated internally is extracted in stead of the perimeter. In such a case, it suffices if a user enters a point on the boundary not extracted using a mouse or any pointing device. Point **A** on the Figure 7 is an example of entered points. If this fails again, another point on the boundary not extracted should be entered. The NC method finds a route passing through these points, resulting to a correct boundary.



Figure 6: Wrong extracted boundary.



Figure 7: Applied the interactive extraction method.

3. Recognition

The recognition process consists of the feature selection and decision process.

3.1. Feature Selection

3.1.1. Shape Features

Based upon the extracted boundary, we define the gravity center of its internal region and identify the starting point where the boundary crosses the y-axis. Then the length l from the starting point is defined along the boundary. Then the distance d (see Figure 8) from the gravity center to the boundary is measured as the function of l .

The following four features are defined as shape features.

F1) The ratio of the average width over the average height $F1 = l/d$, which is an approximation of the average petal width over the average petal height.

F2) The number of petals, which corresponds to the number of valleys or that of hill tops in the above curve. We assign a number of 'N' representing 'many' in case of more than 7 petals in order to deal with some flowers such as a dandelion which has a large but fluctuated number of petals.

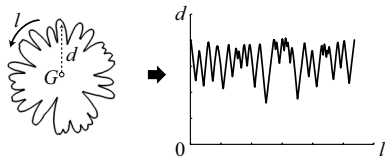


Figure 8: Analysis flower shape.

F3) The central moment is defined as: $F3 = 1/S \sum_x \sum_y ((x - g_x)^2 + (y - g_y)^2)^{1/2}$ where S is the total number of pixels in the flower region, (g_x, g_y) is the center of gravity of the flower region and (x, y) is in the flower region pixel.

F4) The roundness, which is an indicator for how much the perimeter shape is closer to a circle, defined as: $F4 = 4\pi S/L^2$, where S is the same definition of F3 and L is the perimeter of flower region. We have $0 < F4 < 1$.

3.1.2. Color Features

The camera image is given in the RGB color coordinate. We convert the RGB coordinate into the HSV (hue, saturation and value) coordinate[9]. The HS space is divided into 72 cells where each cell is assigned an identical area (Figure 9). For each flower image, its distribution is computed. Then identify the segment which has the largest entry. The selected features are the following six:

- F5) the x coordinate of the largest color cell.
- F6) the y coordinate of the largest color cell.
- F7) the ratio of the largest color cell.
- F8) the x coordinate of the second largest color cell.
- F9) the y coordinate of the second largest color cell.
- F10) the ratio of the second largest color cell.

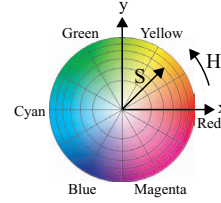


Figure 9: Separate HS space into 12×6 cells.

3.2. Recognition

A piecewise linear discriminant function is used for recognition. For each feature, the observed values are normalized into a value range $[0, 1]$ by using the maximum and minimum values. For estimating a recognition rate of our method, a leave-one-out method is applied.

4. Experiments

First, experiments were conducted for determining how pictures should be taken for focusing an object and defocusing the background. Several pictures were taken by varying F values of a camera (OLYMPUS C-2100). The NC methods extracted some background objects when F value exceeds a certain value. A successful case is shown in Figure 10(a) (with $F=2.8$) while Figure 10(b) (with $F=8.0$) is an unsuccessful case. Note here that the flower (African violet) has a diameter of about 4cm. The distance to the object flower was 10 cm while that to the flower behind the object flower was 14cm.

Pictures were taken with auto focus mode with setting F value from F2.8 to F3.5. Since we have done nothing special, ordinary users are surely able to provide photographs similar to those used here.

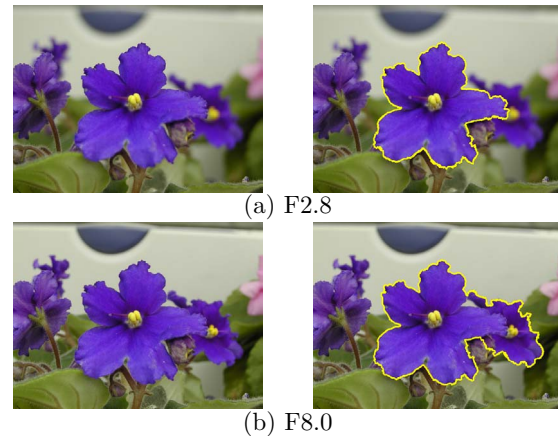


Figure 10: Sample photographs.

We gathered 20 sets of flower picture from 30 species in the fields near our campus.

For boundary extraction, the automatic extraction method described in 2.1 was applied to these 600 pic-

tures. The success extraction rate was 97.2%. The average processing time was 6.69 seconds on a PC (CPU: Pentium IV 3.0GHz with 1GB main memory). Among 17 unsuccessful pictures, three are excessive (i.e. additional perimeters on the background are included) and 14 are under (i.e. internal boundaries are included). For these unsuccessful extractions, the manual method described in 2.2 was applied. The average time for manual operation was less than 10 seconds.

The recognition process was carried out for the 600 photographs with all the 10 features. Out of 20 samples for each species, 19 samples are a training set and the remaining one sample is a recognition set. By varying one sample, the total number of recognition trials is 20 for each species. The resulting recognition rate was 90.7%, 97.7%, and 99.0% if the correct one is included in the top one, top two, and top three candidates, respectively.

The next experiments were to determine which features among ten features are more effective for recognition. First only one feature was used as the recognition feature. The feature yielding the highest recognition rate was identified and then a second feature with the first feature yielding the highest recognition rate was determined. This process was carried out for all the 10 features. The most to least effective features are F4, F5, F2, F7, F1, and F6 in this order. Figure 11 shows the result, where three curves (1st, 2nd, and 3rd) are shown. Six to seven features are required for the top one candidate case (which is the most appropriate for recognition).

We built a recognition system written in C++ language on a PC. Figure 12 shows the main interface menu where the left window shows the input photograph and the resulting extracted boundary for verification. The right window shows the first, second, and the third most likely species.

5. Conclusion

This paper has made two major contributes. The first is the automatic boundary extraction method based upon the NC (normalized cost) method. An additional scheme for dealing with a small percentage of failures is also important. The second is identification of features for flower recognition.

A recognition system was completed. Based on 600 photographs, the extraction rate was 97% with the average processing time of 6.7 seconds. The final recognition rate was 91% with 10 shape and color features.

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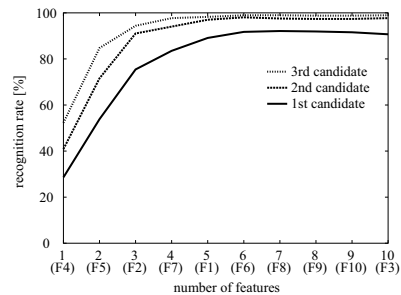


Figure 11: Recognition results.

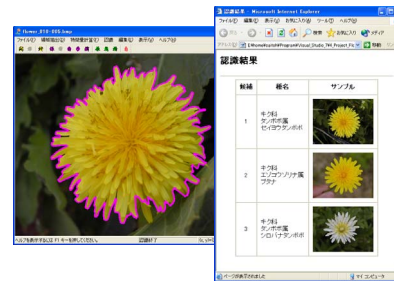


Figure 12: Application and result window.

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